

»» Chapter X ««

SUPPLEMENTARY DEVELOPMENTS CONCERNING $\mathbf{C}$

PREFATORY REMARKS

The questions considered here will be treated largely in exercise form. Use will be made of the first four chapters.

1. OTHER CONSTRUCTIONS OF $\mathbf{C}$

Intended for readers familiar with rings of polynomials and elementary matrix operations.

EXERCISES

1.1. Show that if $\mathbf{R}[x]$ is the ring of polynomials in one indeterminate over $\mathbf{R}$, then $\mathbf{R}[x] \bmod(1 + x^2)$ is $\mathbf{C}$ in the sense that it is a field containing a subfield $\mathbf{R}_0$ which is ordered and complete, that it possesses a member i , $i^2 = -1$, and that it is generated by $\mathbf{R}_0 \cup \{i\}$.

1.2. Let M denote the set of matrices

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix},$$

$a, b \in \mathbf{R}$. Show that M together with matrix addition and multiplication of its members constitutes a field. Show that $\mathbf{R}_0$, the set of matrices of the form

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix},$$

is isomorphic to $\mathbf{R}$, that

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = -\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and that M is generated by the union of $\mathbf{R}_0$ and the singleton whose member is

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

2. CHARACTERIZATIONS OF ABSOLUTE VALUE

The property stated in the first sentence of each of these exercises characterizes absolute value.

EXERCISES

2.1. There is a unique $f: \mathbf{C} \rightarrow \mathbf{R}$ satisfying the following conditions:

- (a) $f(x) = x, \quad x \in \mathbf{R}^+,$
- (b) $f(z + w) \leq f(z) + f(w), \quad f(zw) = f(z)f(w), \quad z, w \in \mathbf{C}.$

Hint: Show that $f(z) > 0, z \neq 0$, and that $f(u) \leq 2, |u| = 1$, and conclude that $f(u) = 1, |u| = 1$.

2.2. There is a unique continuous map f of $\mathbf{C}$ into $\mathbf{R}$ satisfying

- (a) $f(x) = x, \quad x \in \mathbf{R}^+,$
- (b) $f(zw) = f(z)f(w), \quad z, w \in \mathbf{C}.$

Show with the aid of Hamel's discontinuous $g: \mathbf{R} \rightarrow \mathbf{R}$ and satisfying

$$g(x + y) = g(x) + g(y), \quad x, y \in \mathbf{R},$$

that the preceding assertion is false when "continuous" is omitted.

Show that $f: \mathbf{C} - \{0\} \rightarrow \mathbf{R}$ and satisfying (b) is either continuous on $\mathbf{C} - \{0\}$ or continuous at no point of $\mathbf{C} - \{0\}$. Determine the continuous f .

3. GEOMETRY IN $\mathbf{C}$

We recall that $f: X \rightarrow Y$, where X and Y are metric spaces, is termed an *isometry* when it preserves distance, that is, $\rho(a, b) = \sigma[f(a), f(b)]$, $a, b \in X$, where ρ is the metric on X and σ is the metric on Y . We seek the isometries of $\mathbf{C}$ into itself. We know that $uz + a$ where $|u| = 1$ defines an isometry of $\mathbf{C}$

onto itself. Suppose that f is an isometry of $\mathbb{C}$ into itself. Then

$$g(z) = \frac{z - f(0)}{f(1) - f(0)}$$

with domain $\mathbb{C}$ is an isometry. Let $h = g \circ f$. We see that h is an isometry and that $h(0) = 0$, $h(1) = 1$. From $|h(z)|^2 = |z|^2$ and $|h(z) - 1|^2 = |z - 1|^2$, we conclude that $\operatorname{Re} h(z) = \operatorname{Re} z$. Consequently $h(i) = i$ or $h(i) = -i$. In the former case we have $\operatorname{Re}[h(iz)/i] = \operatorname{Re} z$, so that $\operatorname{Im} h(z) = \operatorname{Im} z$ and hence $h(z) = z$, $z \in \mathbb{C}$. In the latter case $\operatorname{Re}[-h(iz)/i] = \operatorname{Re} z$, so that $\operatorname{Im} h(z) = -\operatorname{Im} z$ and hence $h(z) = \bar{z}$, $z \in \mathbb{C}$. The following theorem is now immediate.

Theorem 3.1: *The isometries of $\mathbb{C}$ into itself consist precisely of the maps f of one of the forms $f(z) = uz + a$, $f(z) = u\bar{z} + a$, $a \in \mathbb{C}$, $u \in C(0; 1)$. They all map $\mathbb{C}$ onto itself and constitute a group under the operation of composition.*

Rigid motions. The isometries of $\mathbb{C}$ onto itself of the form $f(z) = uz + a$ are termed *rigid motions*. They constitute a subgroup of the full group of isometries of $\mathbb{C}$ onto itself. A rigid motion f is characterized by the property that f is an isometry and there exists a continuous map F of $\mathbb{C} \times [0, 1]$ into $\mathbb{C}$ such that for each $t \in [0, 1]$ the map taking z into $F(z, t)$ is an isometry and further $F(z, 0) = z$, $F(z, 1) = f(z)$, $z \in \mathbb{C}$. Indeed, if f is a rigid motion so that $f(z) = e^{i\theta}z + a$ for some real θ , then we have an allowed deformation in $F(z, t) = e^{it\theta}z + at$. On the other hand if such an F exists, then for each $t \in [0, 1]$ the map

$$g_t(z) = \frac{F(z, t) - F(0, t)}{F(1, t) - F(0, t)}$$

with domain $\mathbb{C}$ is an isometry keeping 0 and 1 fixed. By continuity $g_t(i) = i$, $0 \leq t \leq 1$, the segment $[0, 1]$ being connected. Hence each g_t is the identity map. The assertion follows.

Angle. For the study of conformal maps we shall want an adequate arithmetic treatment of the notion of angle. It will be defined in terms of rays. Referring to the definition given in Ch. II of a ray, we verify that a ray admits a unique representation of the form $\{a + ut : t \in \mathbb{R}^+\}$, $|u| = 1$. The possibility follows from the definition since

$$\{a + ct : t \in \mathbb{R}^+\} = \{a + ut : t \in \mathbb{R}^+\}, \quad c \neq 0, \quad u = c/|c|.$$

Uniqueness. If

$$\{a + ut : t \in \mathbb{R}^+\} = \{b + vt : t \in \mathbb{R}^+\},$$

then $v^{-1}[(a + uz) - b]$ defines a rigid motion mapping $\mathbf{R}^+$ onto itself. Since $v^{-1}u + v^{-1}(a - b)t^{-1} \geq 0$ for t positive, on taking the limit at $+\infty$ we see that $v^{-1}u \in \mathbf{R}^+$ and that hence $u = v$. Since $t + v^{-1}(a - b) \in \mathbf{R}^+$ for $t \in \mathbf{R}^+$, we conclude that $v^{-1}(a - b) \geq 0$. But if $v^{-1}(a - b) > 0$, then $t + v^{-1}(a - b) > 0$ for all $t \in \mathbf{R}^+$. This is impossible. Hence we are led to the conclusion that $a = b$. Uniqueness is thereby established. We term a the *initial point* and u the *unit direction number* of the ray $\{a + ut: t \in \mathbf{R}^+\}$. It is immediate that an isometry of $\mathbf{C}$ onto itself maps a ray onto a ray.

By an *angle* is meant an ordered pair (l, m) of rays with the same initial point. We ask for a numerical complete *invariant* of an angle with respect to rigid motions; that is, we seek a map A from the set of angles into $\mathbf{C}$ satisfying: (1) $A(f(l), f(m)) = A(l, m)$, f a rigid motion, (2) $A(l_1, m_1) = A(l, m)$ implies that there exists a rigid motion f such that $l_1 = f(l)$, $m_1 = f(m)$.[†] Let the rays l, m have a common initial point a . Let u, v be the unit direction numbers of the rays l, m , respectively. Let $f(z) = c + wz$ be a rigid motion. Then wu and wv are the unit direction numbers of $f(l)$ and $f(m)$, respectively, and we see that $(wv)/(wu) = v/u$. Hence the map A defined by $A(l, m) = v/u$ is a numerical invariant of the equivalence induced in the set of angles by the group of rigid motions. We now verify (2), the fact that A so defined is complete. There exists a unique rigid motion f satisfying $f(l) = l_1$. Indeed it is $h \circ g^{-1}$ where g is the rigid motion mapping $\mathbf{R}^+$ onto l and h is the rigid motion mapping $\mathbf{R}^+$ onto l_1 . With u_1, v_1 the unit direction numbers for l_1, m_1 , respectively, we have $u_1 = wu$ where w is the coefficient of z for f , and hence $v_1 = wv$. Since $f(m)$ is the ray with initial point $f(a)$ and unit direction wv , $m_1 = f(m)$. The condition (2) is fulfilled.

We say, given the angle (l, m) , that m makes an angle α with l provided that α is an argument of $A(l, m)$.

The following exercises treat questions having an elementary geometric character. They are concerned with angle, geometric interpretations of addition and multiplication, maps “linear in z and $\bar{z}$,” and the determination of the discrete subgroups of $(\mathbf{C}, +)$. The last question is intimately connected with the theory of periodic and doubly periodic functions. The maps “linear in z

[†] The whole matter is best formulated in terms of equivalence relations. Given a set X , $E \subset X \times X$ is called an *equivalence relation* (on X) provided that (1) $(x, x) \in E$, $x \in X$, (2) if $(x, y) \in E$, then $(y, x) \in E$, (3) if $(x, y), (y, z) \in E$, then $(x, z) \in E$. Let $C(x) = \{y: (y, x) \in E\}$. The sets $C(x)$ are mutually disjoint or else equal. Their union is X . By an *invariant* of E we shall understand a map f with domain X satisfying: if $(x, y) \in E$, then $f(x) = f(y)$. This is equivalent to saying that $f = g \circ C$ where C is the map taking x into $C(x)$ and g is a map whose domain is the set of $C(x)$. The invariant f is termed *complete* provided that if $f(x) = f(y)$, $(x, y) \in E$. The invariant f is called *numerical* provided that $f: X \rightarrow \mathbf{C}$. In the discussion under way X is the set of angles and E is the set of ordered pairs of angles $((l, m), (l_1, m_1))$ such that there exists a rigid motion f satisfying $l_1 = f(l)$, $m_1 = f(m)$. It is a straightforward matter to check that E is an equivalence relation on X .

and $\bar{z}''$ are just a translation of the complex-valued linear functions with domain $\mathbb{R}^2$ into complex terms via the canonical map σ of Ch. II. Their study paves the way for the introduction of complex differential coefficients in Ch. XI.

EXERCISES

3.1. $A(f(l), f(m)) = A(l, m)$ when $f(z) = a + bz$, and $A(f(l), f(m)) = \overline{A(l, m)}$ when $f(z) = a + b\bar{z}$. In both cases it is understood that $b \neq 0$.

3.2. If $l = \mathbb{R}^+$ and $m = \{e^{i\theta}t: t \in \mathbb{R}^+\}$ where $\theta \in \mathbb{R}$, then m makes an angle θ with l .

3.3. Every rigid motion is a rotation or a translation. Every rigid motion not a translation admits a representation of the form $S \circ T$ where S is a rotation with fixed point 0 and T is a translation. State and prove an analog for a representation of the form $T \circ S$ (cf. §2, Ch. II).

3.4. Geometric interpretation of multiplication. Given $c \in \mathbb{C} - \{0\}$, let l_c denote the ray with initial point 0 which passes through c . Now suppose that $a, b \in \mathbb{C} - \{0\}$. Then $A(l_b, l_{ba}) = A(l_1, l_a)$ and $|ba| = |a| |b|$. Thus ba lies on the ray $f(l_a)$ where f is the unique rigid motion satisfying $f(l_1) = l_b$; actually $f(z) = (b/|b|)z$. We have $ba = g \circ f(a)$, where $g(z) = |b|z$, a homothety. Also

$$(0, b, ba) = (g \circ f(0), g \circ f(1), g \circ f(a))$$

shares 0, b with $(0, 1, b)$.

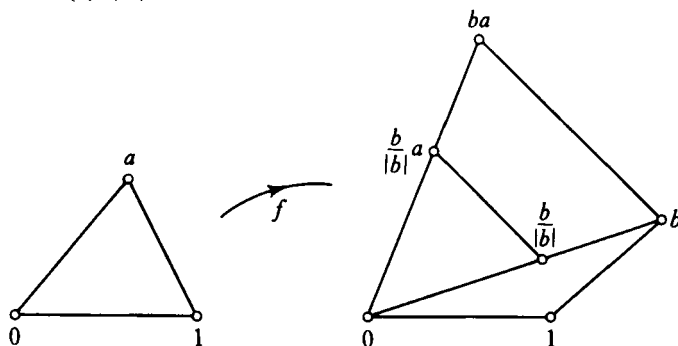


FIG. 10.1

3.5. Given $z_1, z_2, z_3 \in \mathbb{C}$ and $z_j \neq z_k, j \neq k$, a necessary and sufficient condition that $w_k = f(z_k), k = 1, 2, 3$, for some $f(z) = az + b, a \neq 0$, is that $w_j \neq w_k, j \neq k$ and

$$\frac{w_3 - w_1}{w_2 - w_1} = \frac{z_3 - z_1}{z_2 - z_1}.$$

Treat the analogous problem for $f(z)$ of the form $a\bar{z} + b$, and also for rigid motions and isometries. Rephrase the results, as far as possible, in the language of invariants. What is the situation for two-point problems of this kind?

3.6. Given $a, b, c \in \mathbb{C}$, let $f(x, y) = ax + by + c$ with domain $\mathbb{R}^2$. Show that f is univalent if and only if $\text{Im}(b\bar{a}) \neq 0$ and also that f is univalent if and only if $f(\mathbb{R}^2) = \mathbb{C}$.

3.7. Recall the canonical map σ of Ch. II, $\sigma(x, y) = x + yi$. Show that the maps $f \circ \sigma^{-1}, f$ as in Ex. 3.6, consist precisely of the maps $g(z) = \alpha z + \beta \bar{z} + \gamma$, $\alpha, \beta, \gamma \in \mathbb{C}$, with domain $\mathbb{C}$. Show directly without reference to Ex. 3.6 that $g(z) = 0$, $z \in \mathbb{C}$, if and only if $\alpha = \beta = \gamma = 0$. Show that g is univalent if and only if $|\alpha|^2 - |\beta|^2 \neq 0$ and also that g is univalent if and only if $g(\mathbb{C}) = \mathbb{C}$. Show that the set of univalent g constitutes a group under composition and determine the coefficients associated with g^{-1} explicitly.

For the special case where $\gamma = 0$ determine

$$[\max_{C(0;1)} |g|][\min_{C(0;1)} |g|].$$

[When $|\alpha|^2 - |\beta|^2 \neq 0$, $g[C(0; 1)]$ is an ellipse.]

3.8. Determine g of Ex. 3.7 satisfying $g[g(z)] = z$, $z \in \mathbb{C}$. They are the so-called linear *involutions* of $\mathbb{C}$.

3.9. Let l be an isometry of $\mathbb{C}$ onto itself. By an l -reflexion is meant a $\varphi: \mathbb{C} \rightarrow \mathbb{C}$ such that $\varphi \circ l(z) = l(\bar{z})$, $z \in \mathbb{C}$. Note that each point of $l(\mathbb{R})$ is a fixed point of φ . Show that $\varphi(z) = f^{-1}[\overline{f(z)}]$ where f is a rigid motion. Show that if m is an isometry of $\mathbb{C}$ onto itself satisfying $m(\mathbb{R}) = l(\mathbb{R})$, it determines the same φ as l .

3.10. Show that the straight lines are precisely the sets of the form $\{z: \text{Im}(\bar{u}z) = c\}$, $u \in C(0; 1)$, $c \in \mathbb{R}$. Show that u is determined up to a factor of -1 . Show that two lines are equal or disjoint (that is, are parallel) if and only if both have the same associated u .

3.11. Geometric interpretation of addition. Nondegenerate case. Given $a, b \in \mathbb{C}$, $\text{Im}(b\bar{a}) \neq 0$. Show that $a + b$ is the point of intersection of the line l containing b and parallel to the line containing 0 and a and of the line m containing a and parallel to the line containing 0 and b . What can be said about the degenerate cases?

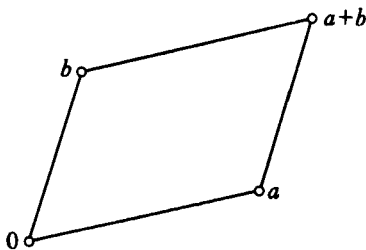


FIG. 10.2

3.12. Given parallel lines $\{\text{Im}(\bar{u}z) = A\}$ and $\{\text{Im}(\bar{u}z) = B\}$, $|u| = 1$. By their associated strip is meant the set

$$\{\min\{A, B\} \leq \text{Im}(\bar{u}z) \leq \max\{A, B\}\}.$$

Note that the frontier (in the sense of C) of the strip is just the union of the two given lines. By the *parallelogram determined by the ordered triple* (a, b, c) , $\text{Im}(c\bar{b}) \neq 0$, is meant the intersection of the strip associated with $\{\text{Im}[\bar{b}(z - a)] = 0\}$ and $\{\text{Im}[\bar{b}(z - a)] = \text{Im}(\bar{b}c)\}$ and the strip associated with $\{\text{Im}[\bar{c}(z - a)] = 0\}$ and

$\{\operatorname{Im}[\bar{c}(z - a)] = \operatorname{Im}(\bar{c}b)\}$. Show that the parallelogram determined by (a, b, c) is the image of $[0, 1] \times [0, 1]$ with respect to $a + bx + cy$.

3.13. A *discrete* topological space is one where each singleton is open. A subset of a topological space is termed a discrete subset provided that it is a discrete topological space in the sense of the relative topology. Show that the discrete subgroups of $(\mathbb{C}, +)$ consist of: (1) the trivial one reducing to 0, (2) the subgroups generated by a single element $a \neq 0$, (3) the subgroups generated by a pair $\{a, b\}$, $\operatorname{Im}(b\bar{a}) \neq 0$.

This may be carried out as follows. First verify that each of the listed subgroups is discrete. In the reverse direction we note that the group property implies that the underlying set of a discrete subgroup of $(\mathbb{C}, +)$ clusters at no point of $\mathbb{C}$. In case $G \neq \{0\}$ is such a subgroup we introduce $a \in G$ satisfying $|a| = \min\{|z| : z \in G - \{0\}\}$ and consider the possibilities (1) G is a subset of the line containing 0 and a , (2) the supposition (1) is false. For (1) show that $G = \{na : n \in \mathbb{Z}\}$. For (2) take $b \in G$ satisfying $|b| = \min\{|z| : z \in G, \operatorname{Im}(z\bar{a}) \neq 0\}$, verifying the existence of such b . Observe that if $sa + tb \in G$, $0 \leq s, t < 1$, $(s, t) \neq (0, 0)$, then $\sigma a + \tau b \in G$ for some (σ, τ) , $0 < |\sigma|, |\tau| \leq \frac{1}{2}$, and hence $|\sigma a + \tau b| < |b|$. Conclude that G is generated by $\{a, b\}$. Indicate on a diagram the distribution of points generated by $\{a, b\}$ when the second possibility occurs.

3.14. The subgroups of $(\mathbb{C}, +)$ generated by $\{a_k, b_k\}$, $\operatorname{Im}(b_k\bar{a}_k) > 0$, $k = 1, 2$, are equal if and only if there exists a matrix of real integers

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

satisfying $a_2 = \alpha a_1 + \beta b_1$, $b_2 = \gamma a_1 + \delta b_1$, $\alpha\delta - \beta\gamma = 1$. There is at most one matrix with real entries satisfying the first two stated conditions. The set of such matrices constitutes an important group under matrix multiplication (it is a modular group, cf. [47]).

3.15. Determine the discrete subgroups of $(\mathbb{C} - \{0\}, \cdot)$. Hint: The set of all logarithms of the members of such a group is a discrete subgroup of $(\mathbb{C}, +)$ containing the element $2\pi i$.

3.16. Let f be meromorphic on $\mathbb{C}$ and not constant. Show that the set of univalent analytic maps α of $\mathbb{C}$ onto itself satisfying $f \circ \alpha = f$ constitutes a group G . Show that G has the property that for each $z \in \mathbb{C}$ there exists a neighborhood U of z such that $\{\alpha : \alpha \in G, \alpha(z) \in U\}$ is finite.